

Lesson 9

Managing Post-Installation Administration Tasks

Topic 9A

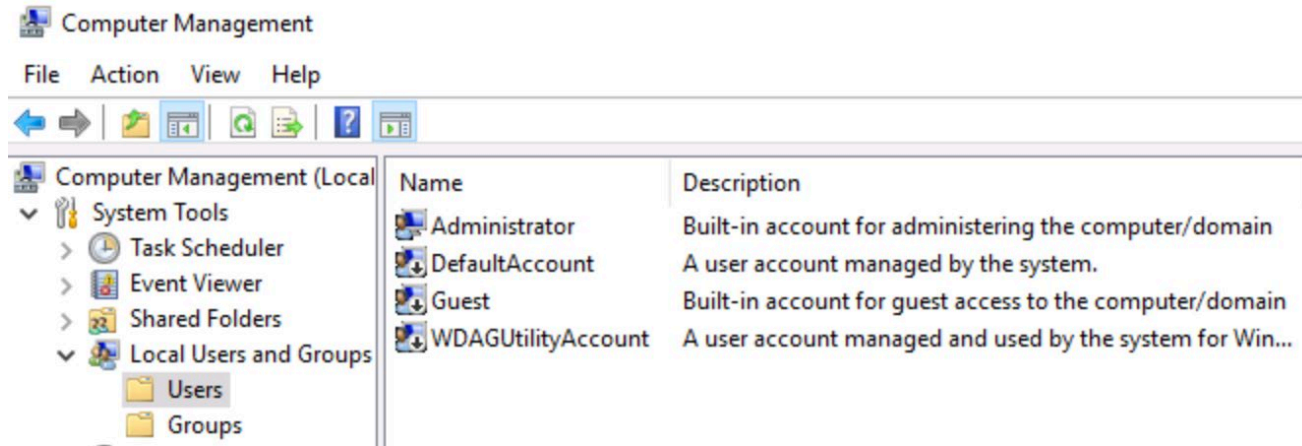
Understand Secure Administration Practices

Exam Objectives Covered

- 3.3 Explain important concepts pertaining to identity and access management for server administration

Managing Users and Groups

- Local user accounts in Windows



The Local Users and Groups console on Windows Server. (Screenshot courtesy of Microsoft.)

Managing Users and Groups (slide 2)

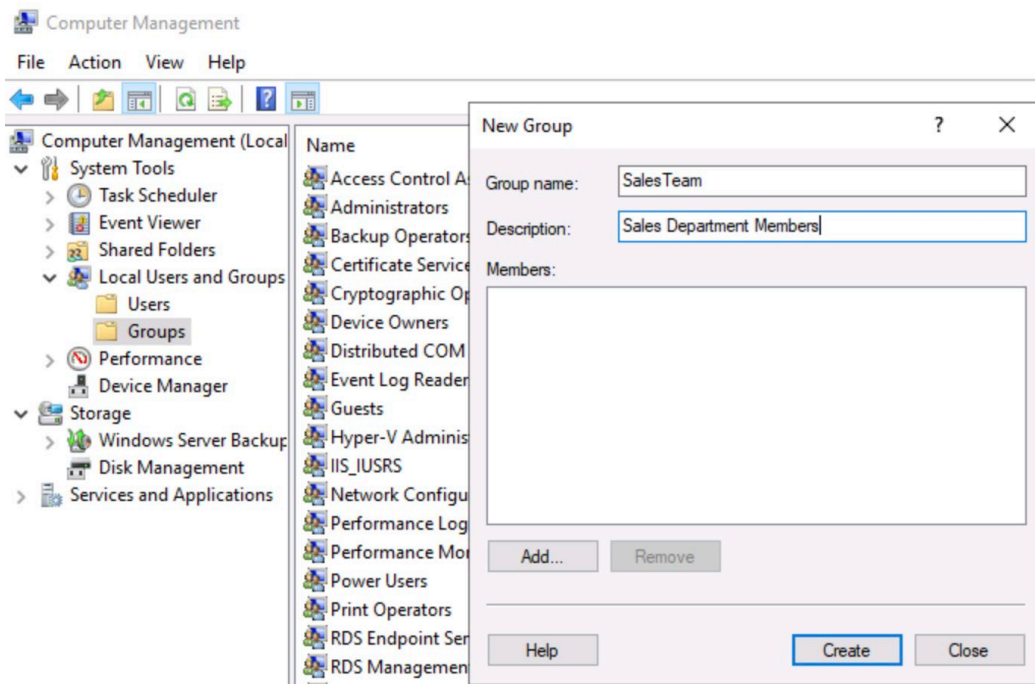
- Local user accounts in Linux

```
[root@localhost ~]# useradd user01  
[root@localhost ~]# █
```

Creating a new local user by using the useradd command in Linux.

Managing Users and Groups (slide 3)

- Local Windows groups



Creating a new local group account in the Local Users and Groups console on Windows Server. (Screenshot courtesy of Microsoft.)

Managing Users and Groups (slide 4)

- Local Linux groups

```
[root@localhost ~]# groupadd sales  
[root@localhost ~]# usermod -aG sales user01  
[root@localhost ~]# █
```

Using the usermod command to add a user to a group.

- Delegation

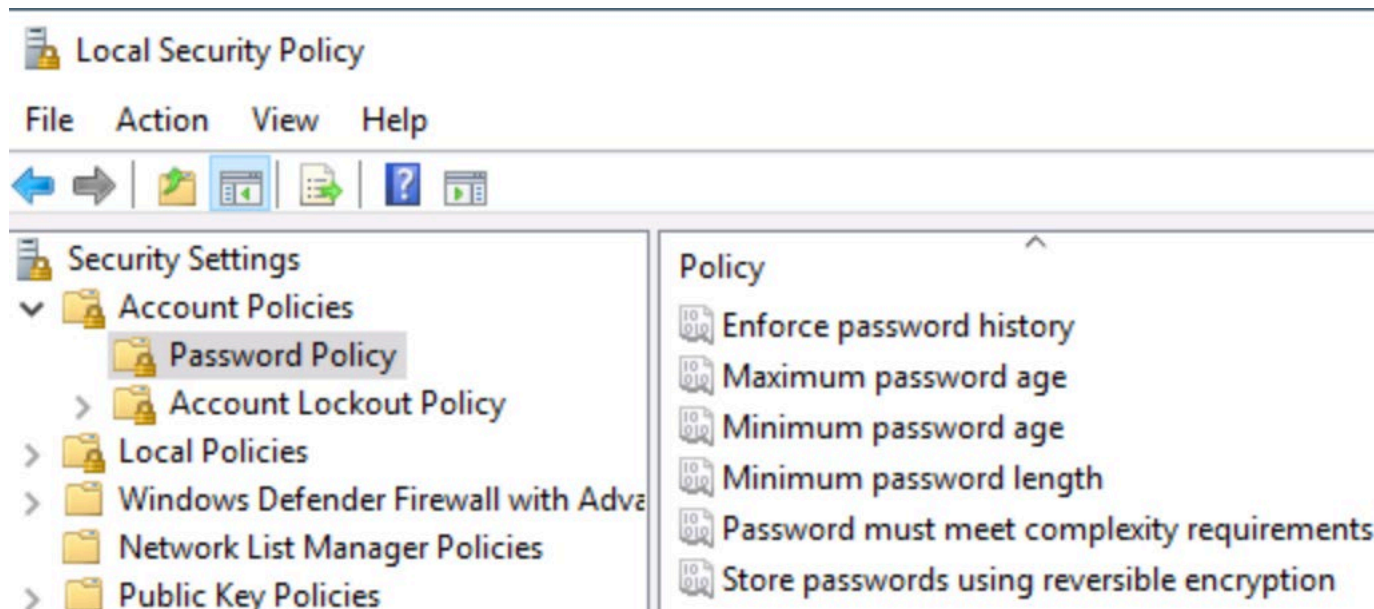
Password Policies

- [illegible]

The Kali Linux penetration testing distribution. (Screenshot courtesy of Wireshark.)

Enforcing Password Policies

- Enforce password policies in Windows



The path in Local Security Policies to configure password settings. (Screenshot courtesy of Microsoft.)

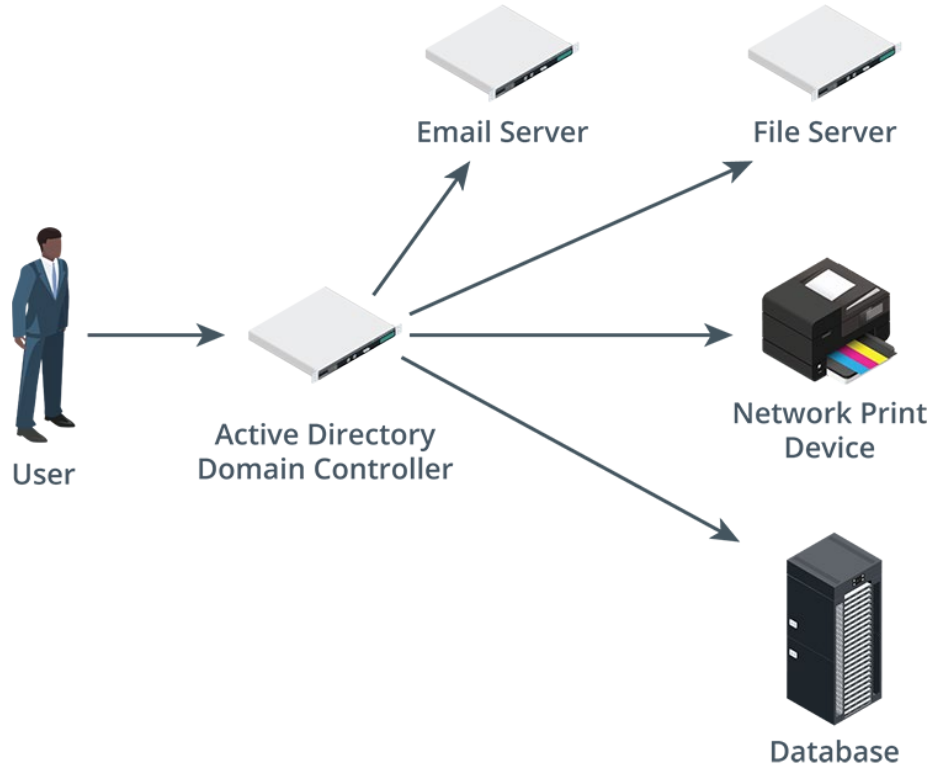
Enforcing Password Policies (slide 2)

- Enforce password policies in Linux
 - PAM configuration file
- Password best practices review

Multifactor Authentication

- What you know
- What you have
- What you are

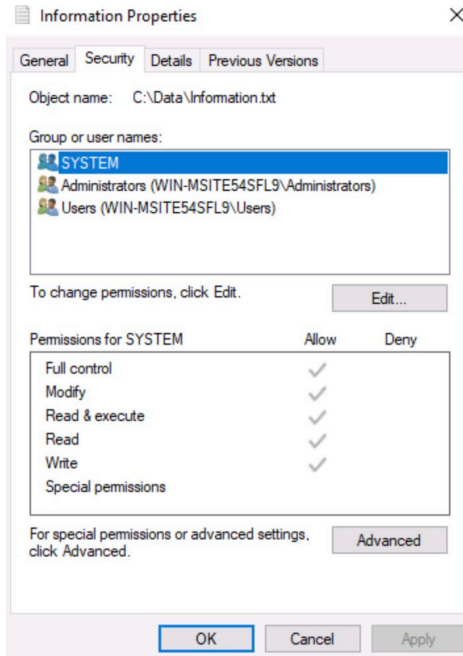
Single Sign-on



After authenticating to AD, the user has access to file, print, email, and database resources. (Images © 123RF.com)

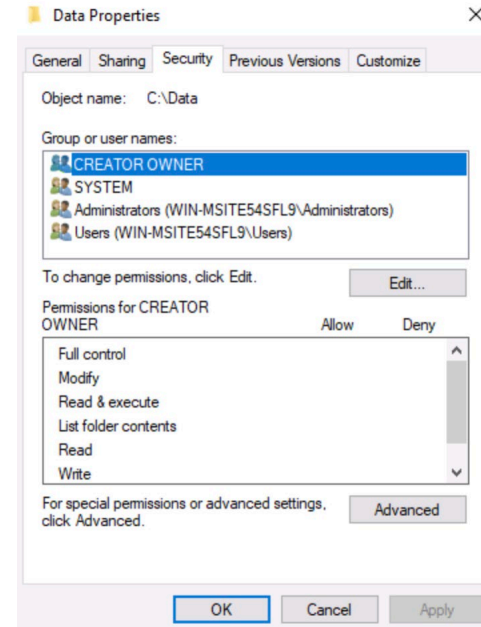
Access Controls

- Windows file security



The Security tab in the Properties of a file. (Screenshot courtesy of Microsoft.)

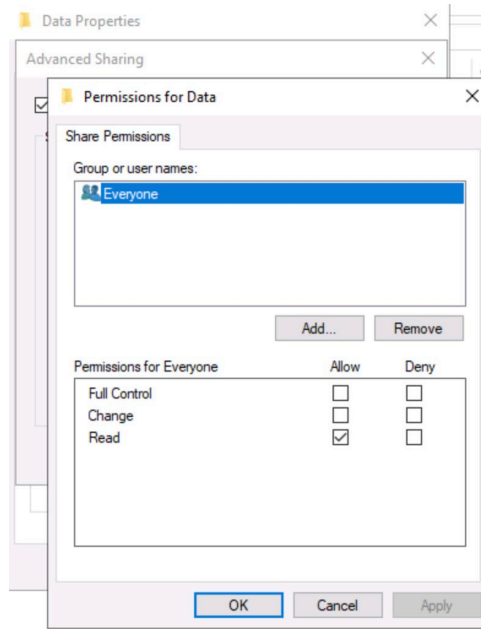
- Windows folder security



The Security tab in the Properties of a folder. (Screenshot courtesy of Microsoft.)

Access Controls (slide 2)

- Windows share permissions



The Security tab in the Properties of a folder. (Screenshot courtesy of Microsoft.)

Access Controls (slide 3)

- Standard Linux permissions

```
[root@localhost ~]# chmod 760 file1.txt
[root@localhost ~]# ls -l file1.txt
-rwxrw----. 1 root root 0 Jan 26 15:00 file1.txt
[root@localhost ~]#
[root@localhost ~]# chmod 440 file1.txt
[root@localhost ~]# ls -l file1.txt
-r--r-----. 1 root root 0 Jan 26 15:00 file1.txt
[root@localhost ~]#
[root@localhost ~]# chmod 644 file1.txt
[root@localhost ~]# ls -l file1.txt
-rw-r--r--. 1 root root 0 Jan 26 15:00 file1.txt
[root@localhost ~]#
```

Examples of using absolute mode to configure permissions.

Access Controls (slide 4)

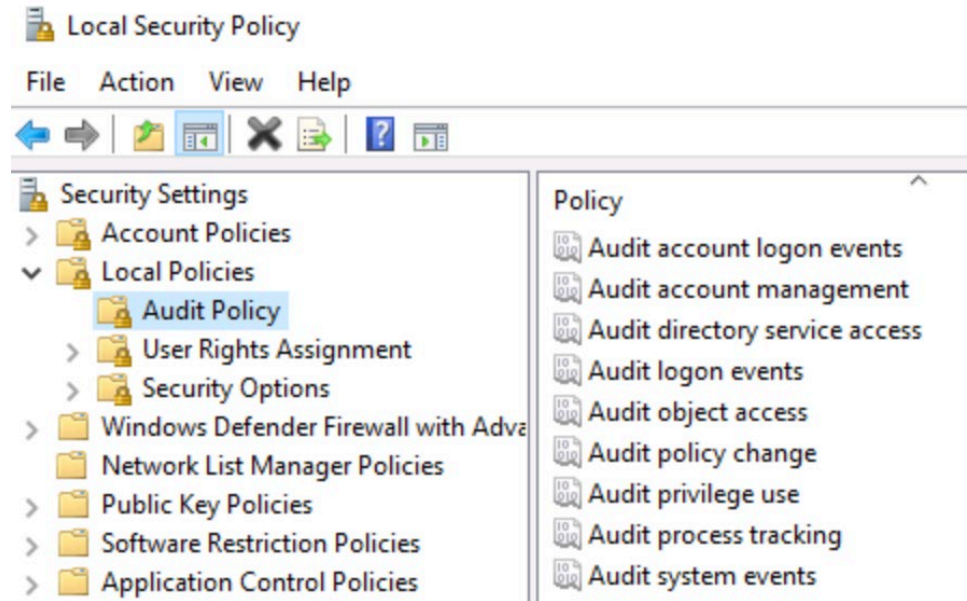
- Standard Linux permissions

```
[root@localhost ~]# chmod u=rwx,g=rw,o= file1.txt
[root@localhost ~]# ls -l file1.txt
-rwxrw----. 1 root root 0 Jan 26 15:00 file1.txt
[root@localhost ~]#
[root@localhost ~]# chmod ug=r,o= file1.txt
[root@localhost ~]# ls -l file1.txt
-r--r-----. 1 root root 0 Jan 26 15:00 file1.txt
[root@localhost ~]#
[root@localhost ~]# chmod u=rw,go=r file1.txt
[root@localhost ~]# ls -l file1.txt
-rw-r--r--. 1 root root 0 Jan 26 15:00 file1.txt
[root@localhost ~]# █
```

Examples of using symbolic mode to configure permissions.

Account Auditing

- User
- Group
- Third-party



Audit policy options in Windows Server. (Screenshot courtesy of Microsoft.)

Guidelines for Secure Administration Practices

- Don't share user accounts
- Use groups
- Review password policies
- Audit passwords
- Use MFA
- Windows uses group policy, NTFS and share permissions
- Linux uses PAM, ACLs, standard and NFS permissions

Review Activity: Secure Administration Practices



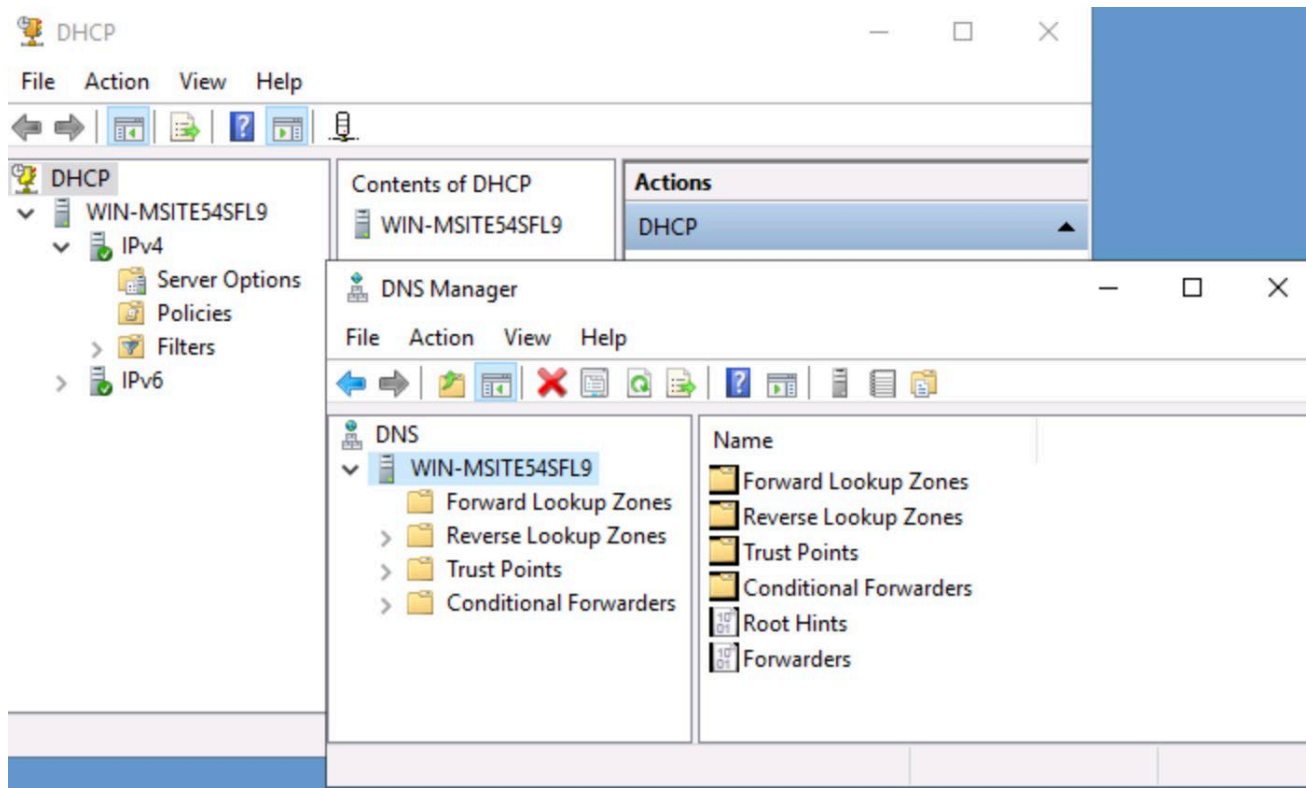
Topic 9B

Manage Server Functions

Exam Objectives Covered

- 2.3 Given a scenario, configure and maintain server functions and features

Server Roles Requirements



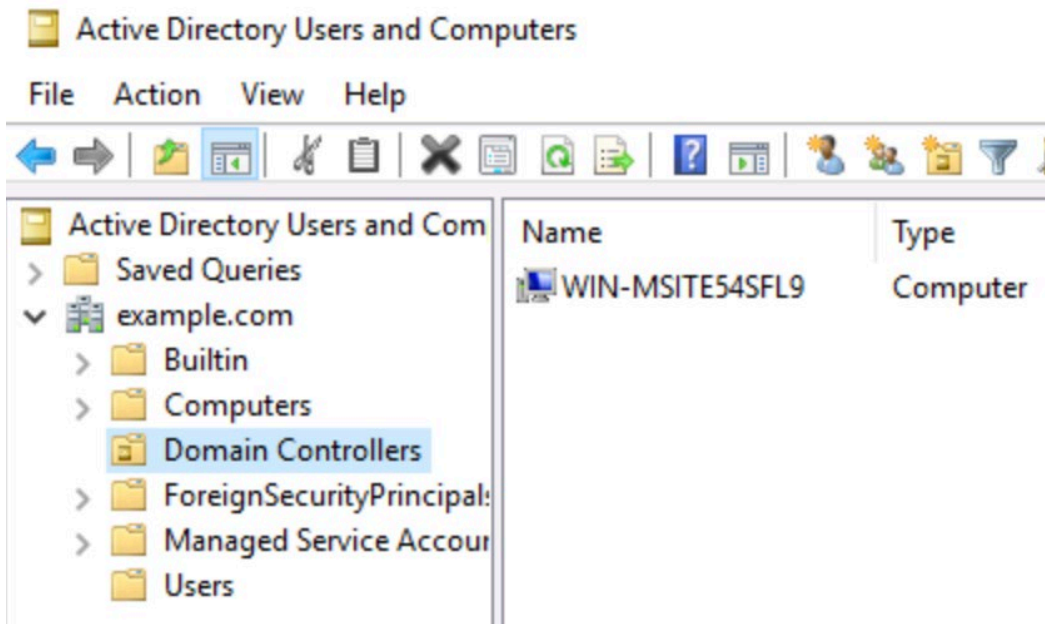
The DNS service and the DHCP service co-located on a single server. (Screenshot courtesy of Microsoft.)

Server Roles Requirements (slide 2)

- General servers
 - File
 - Print
 - Web
- Directory services servers
 - User accounts
 - Computer accounts
- Application servers
 - Database
 - Application/Middleware
 - Messaging/Email
 - Network infrastructure
 - VPN
 - Collaboration
 - VM hosts

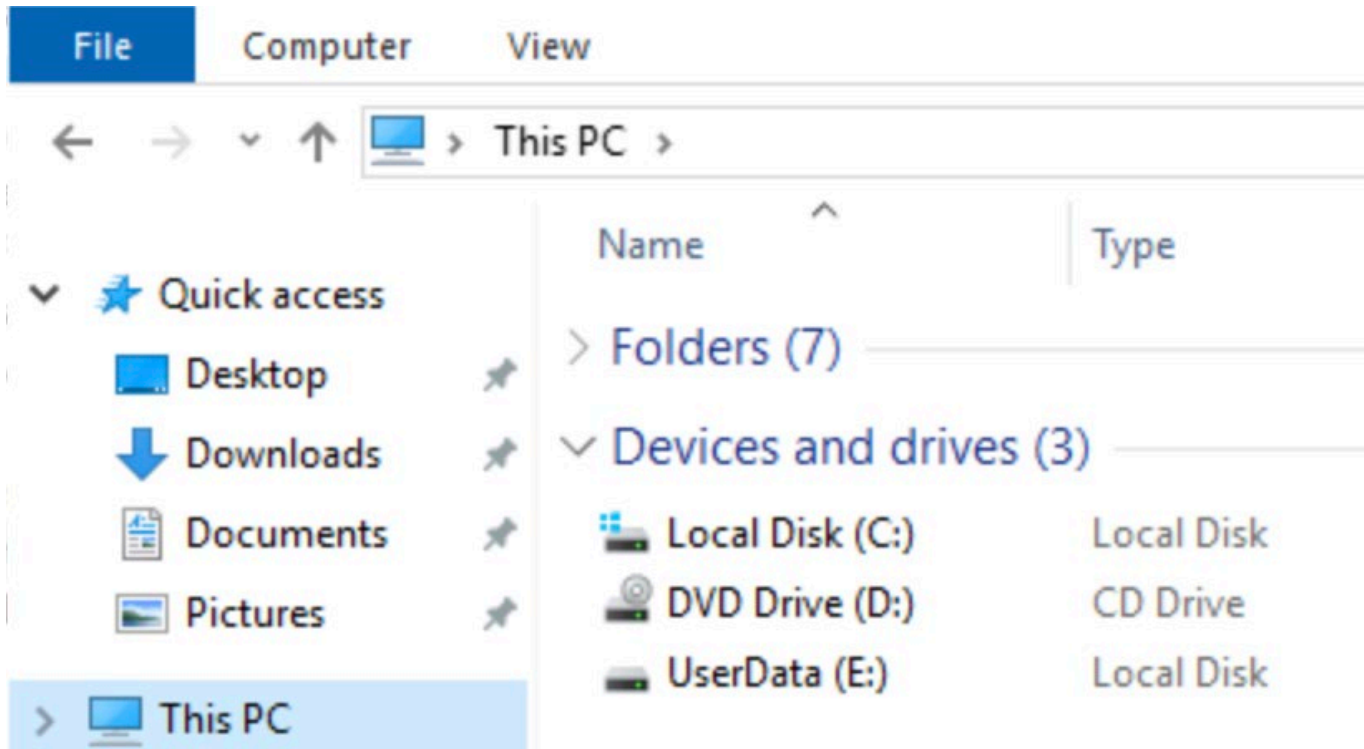
Server Roles Requirements (slide 3)

- Directory services configuration controls



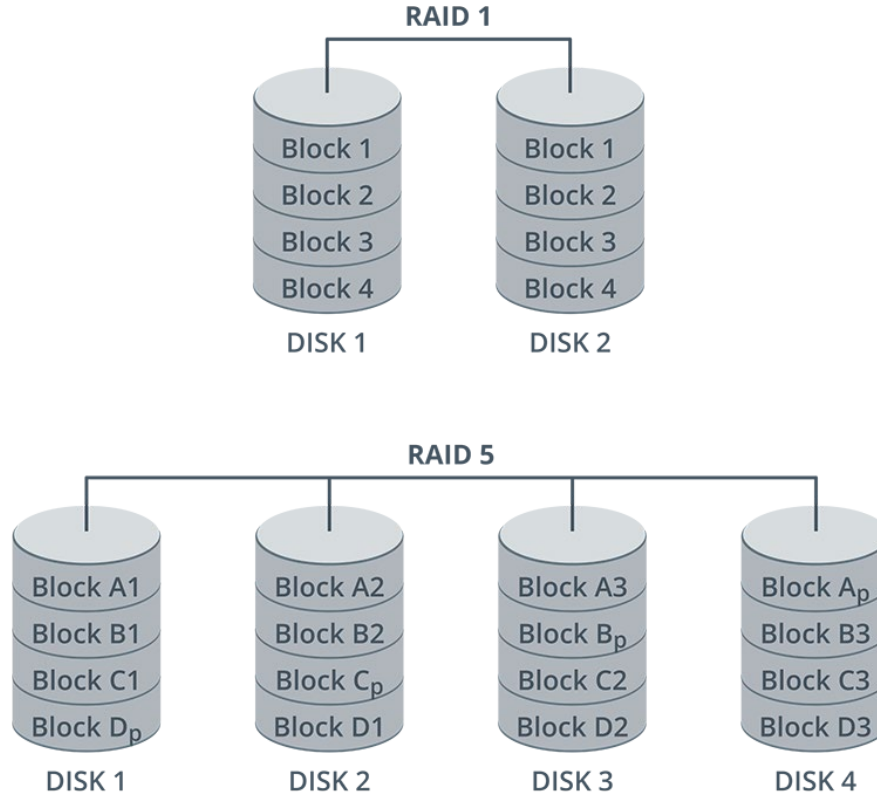
The AD Users and Computers console is used to organize accounts, partly for the purpose of configuration management via Group Policy. (Screenshot courtesy of Microsoft.)

Storage Management



A Windows server with the OS on C: and user data on E: (Screenshot courtesy of Microsoft.)

Storage Management (slide 2)



RAID 1 holds the operating system while RAID 5 holds the user data. (Images © 123RF.com)

Storage Management (slide 3)

- Linux directories

/ - root

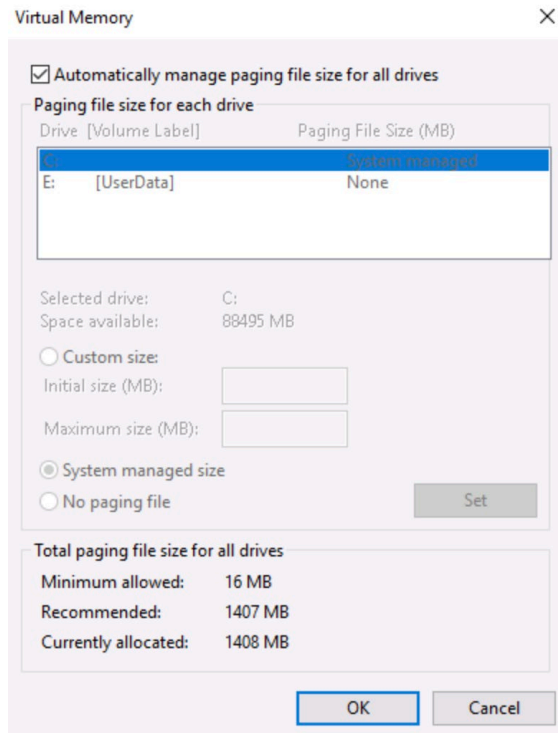
/home

/var

swap

Virtual Memory

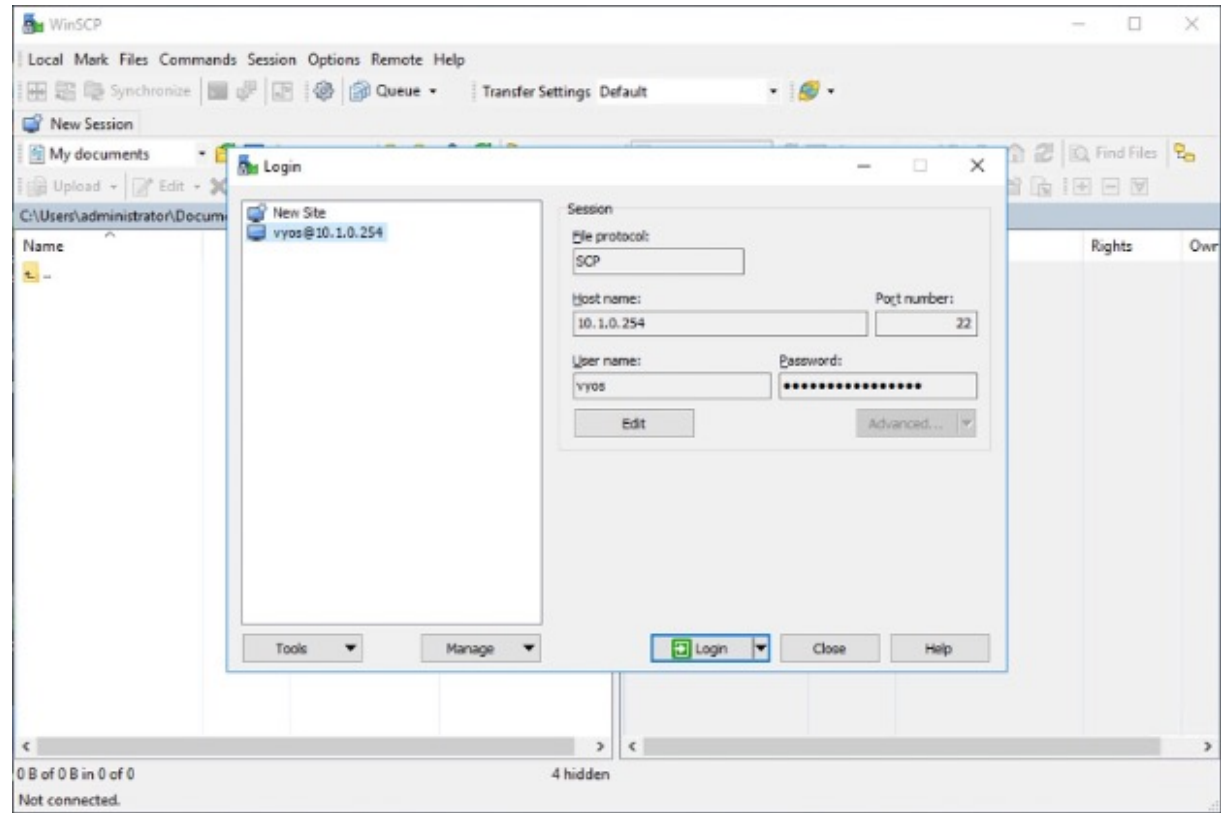
- Recommended settings
 - Ubuntu Linux
 - Windows Server



*Windows Virtual Memory pagefile configuration.
(Screenshot courtesy of Microsoft.)*

Data Transfers

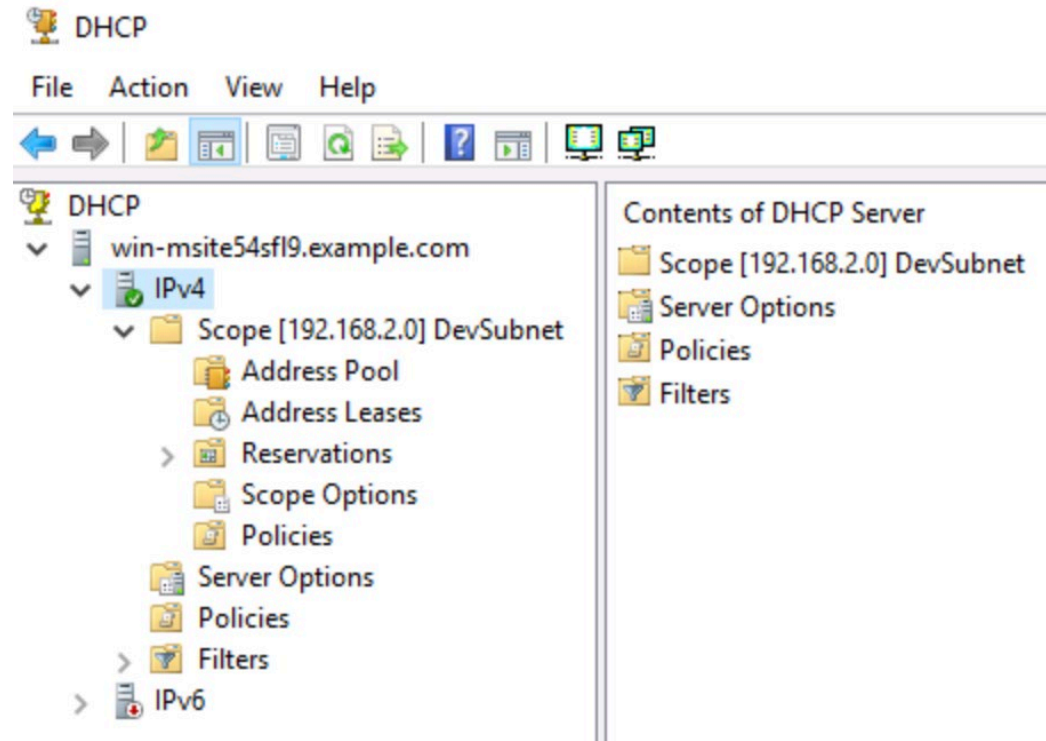
- Administrative tools
- Robocopy
- Rsync
- Scp
- Security breaches and Data exfiltration



WinSCP. (Screenshot courtesy of Microsoft.)

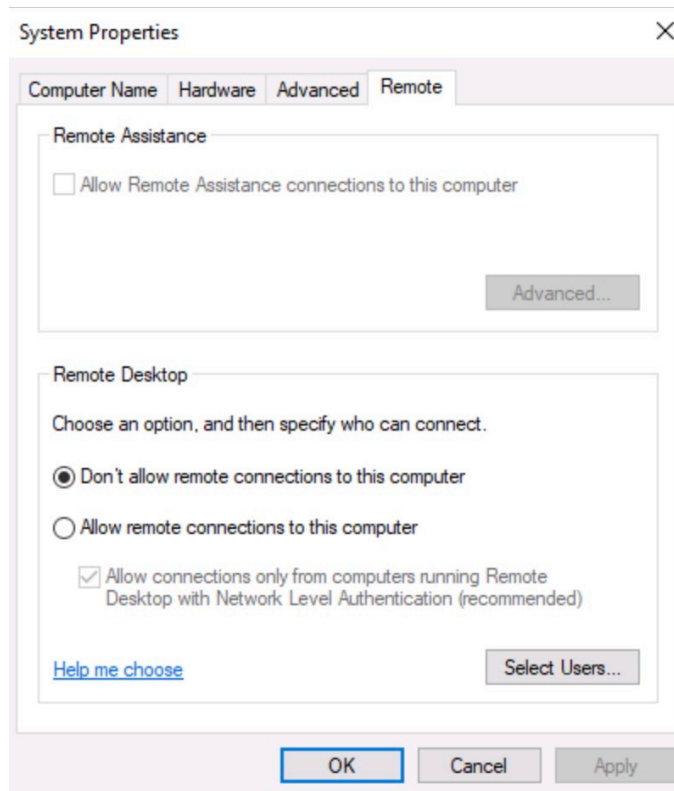
Administrative Interfaces

- Console
- Remote desktop
- SSH
- Web interface



DHCP console in Windows Server. (Screenshot courtesy of Microsoft.)

Administrative Interfaces (slide 2)



Remote Desktop configuration. (Screenshot courtesy of Microsoft.)

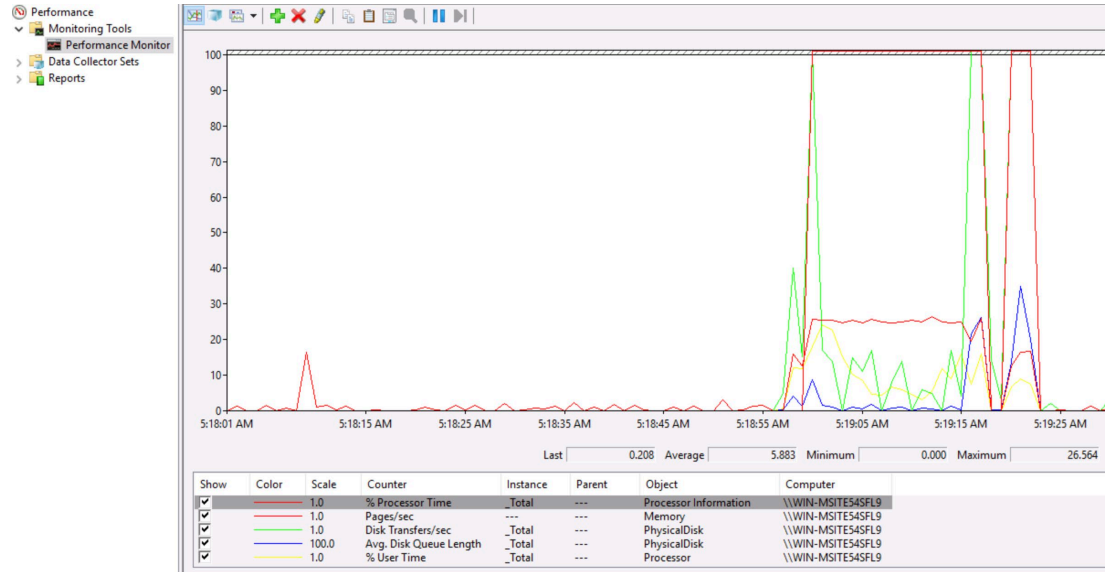
Administrative Interfaces (slide 3)

```
[user01@CentOS7-A ~]$ ssh root@192.168.1.100  
root@192.168.1.100's password:  
Last login: Mon Aug  3 15:15:09 2020 from 172.20.0.2  
[root@CentOS7-A ~]# █
```

SSH connection to 192.168.1.100

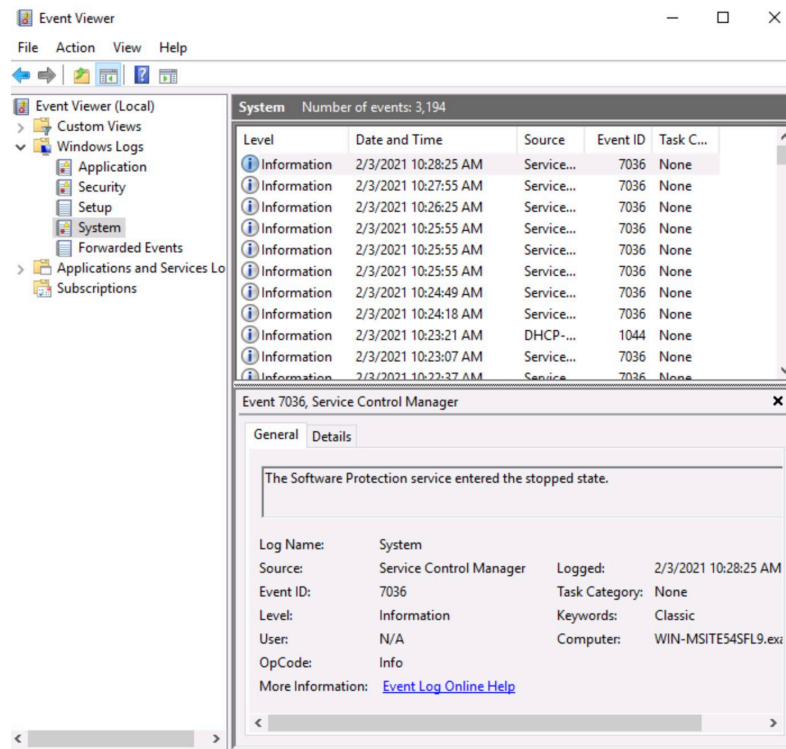
Monitoring

- Windows monitoring tools
- Performance monitor
- Alerts and thresholds
- CPU monitoring
- IOPS
- System uptime



Windows Performance Monitor. (Screenshot courtesy of Microsoft.)

Monitoring (slide 2)



Windows Event Viewer. (Screenshot courtesy of Microsoft.)

Monitoring (slide 3)

- Linux monitoring with Top

```
top - 15:11:45 up 22 min,  1 user,  load average: 0.00, 0.02, 0.07
Tasks: 216 total,   1 running, 215 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.7 us,  0.0 sy,  0.0 ni, 98.3 id,  0.3 wa,  0.7 hi,  0.0 si,  0.0 st
MiB Mem :  3927.9 total,  1917.2 free,   776.6 used,  1234.1 buff/cache
MiB Swap:  1963.0 total,  1963.0 free,    0.0 used.  2906.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1691	damon	20	0	3835420	255780	122460	S	0.7	6.4	0:05.34	gnome-shell
2977	root	20	0	234964	4604	3960	R	0.3	0.1	0:00.02	top
1	root	20	0	174992	17428	11556	S	0.0	0.4	0:01.27	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
.	.	.	.	.	.	.	.	.	.	.	.

Output from the top command.

Monitoring (slide 4)

- Sar
- Linux uptime

```
[root@localhost ~]# uptime
 15:13:06 up 24 min,  1 user,  load average: 0.00, 0.01, 0.06
[root@localhost ~]#
```

Output of the uptime command.

Monitoring (slide 5)

- Linux storage information

```
[root@localhost ~]# du -h /var/log/audit
2.2M    /var/log/audit
[root@localhost ~]#
```

Output of the du command.

```
[root@localhost ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        1.9G   0    1.9G   0% /dev
tmpfs           2.0G   0    2.0G   0% /dev/shm
tmpfs           786M  1.4M   785M   1% /run
/dev/sda2       13G   6.8G   5.7G  55% /
tmpfs           2.0G 392K   2.0G   1% /tmp
/dev/sda2       13G   6.8G   5.7G  55% /home
/dev/sda1       976M  229M   681M  26% /boot
tmpfs           393M  156K   393M   1% /run/user/1000
```

Output of the df command.

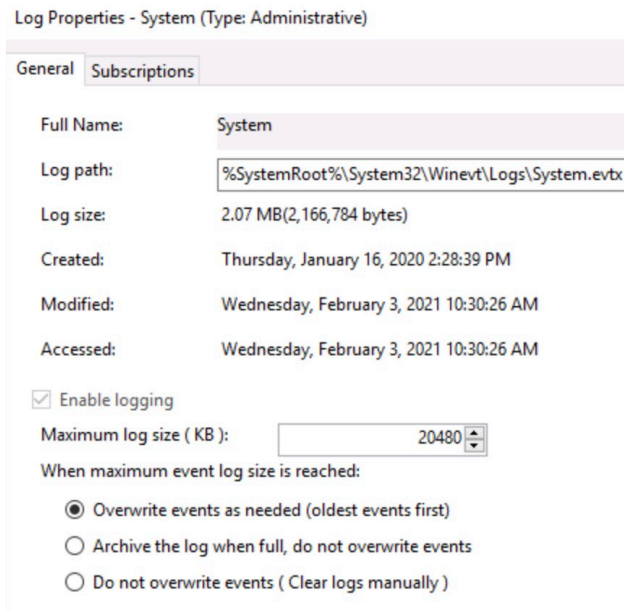
Monitoring (slide 6)

```
[root@localhost log]# head boot.log
    Starting Authorization Manager...
[  OK  ] Finished Rotate log files.
[  OK  ] Started ABRT Automated Bug Reporting Tool.
[  OK  ] Started Creates ABRT problems from coredumpctl messages.
[  OK  ] Started ABRT kernel log watcher.
[  OK  ] Started ABRT Xorg log watcher.
[  OK  ] Started Authorization Manager.
[  OK  ] Finished Builds and install new kernel modules through DKMS.
[  OK  ] Started Disk Manager.
[  OK  ] Started System Security Services Daemon.
[root@localhost log]# █
```

Linux boot log file located in the /var/log directory.

Monitoring (slide 7)

- Log file reporting and retention
- Log shipping



Windows log overwrite screen. (Screenshot courtesy of Microsoft.)

Review Activity: Server Functions



Review Activity

Lesson 9

Summary

